Introduction

Exploratory data analysis (EDA) plays a vital role in the data analysis process, serving as the foundation for discovering meaningful insights, patterns, and trends. Through EDA, we begin to understand the structure and behavior of data, allowing us to make more informed and confident decisions. This project provides a valuable opportunity to apply what I have learned in a practical, hands-on context that closely mirrors real-world analytics work.

By working with a dataset containing Major League Baseball (MLB) player statistics, I will engage in the full EDA cycle—from cleaning and preparing the data, to exploring distributions and relationships, to conducting statistical analysis and drawing conclusions. Each step in the process will help strengthen my ability to think critically about data and refine the techniques I have learned.

In [12]:
```
import pandas as pd
```

In [13]:
```
df = pd.read_csv("mlb.csv")
df.head()
```

Out[13]:

	salary	player_position	race	team_payroll	years_played	games_played	career_at_bats	career_runs	career_hits	career_dou
0	3375000	shortstop	black	38407380	8	918	3333	407	863	
1	3100000	firstbase	white	38407380	5	751	2807	370	840	
2	2900000	thirdbase	white	38407380	8	1056	3337	405	816	
3	1650000	outfield	black	38407380	12	1196	3603	437	928	
4	700000	outfield	black	38407380	17	2032	7489	1136	2145	

5 rows × 21 columns

1. Generate descriptive statistics for the dataset, and comment on the main trends.

In [15]:
```
df.shape
```

Out[15]:
```
(352, 21)
```

In [16]:
```
continuous_variables = df.select_dtypes(include=['float64', 'int64']).columns
continuous_variables.tolist()
```

Out[16]:
```
['salary',
 'team_payroll',
 'years_played',
 'games_played',
 'career_at_bats',
 'career_runs',
 'career_hits',
 'career_doubles',
 'career_triples',
 'career_homeruns',
 'career_runs_batted_in',
 'career_batting_average',
 'career_walks',
 'career_strike_outs',
 'career_stolen_bases',
 'career_fielding_percent',
 'white_percent_in_city',
 'black_percent_in_city',
 'hispanic_percent_in_city']
```

In [17]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 352 entries, 0 to 351
Data columns (total 21 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   salary                    352 non-null    int64
 1   player_position           352 non-null    object
 2   race                      352 non-null    object
 3   team_payroll              352 non-null    int64
 4   years_played              352 non-null    int64
 5   games_played              352 non-null    int64
 6   career_at_bats            352 non-null    int64
 7   career_runs               352 non-null    int64
 8   career_hits               352 non-null    int64
 9   career_doubles            352 non-null    int64
 10  career_triples            352 non-null    int64
 11  career_homeruns           352 non-null    int64
 12  career_runs_batted_in     352 non-null    int64
 13  career_batting_average    352 non-null    int64
 14  career_walks              352 non-null    int64
 15  career_strike_outs        352 non-null    int64
 16  career_stolen_bases       352 non-null    int64
 17  career_fielding_percent   352 non-null    int64
 18  white_percent_in_city     329 non-null    float64
 19  black_percent_in_city     329 non-null    float64
 20  hispanic_percent_in_city  329 non-null    float64
dtypes: float64(3), int64(16), object(2)
memory usage: 57.9+ KB
```

In [18]:
```
df.describe()
```

Out[18]:

	salary	team_payroll	years_played	games_played	career_at_bats	career_runs	career_hits	career_doubles	career
count	3.520000e+02	3.520000e+02	352.000000	352.000000	352.000000	352.000000	352.000000	352.000000	352
mean	1.331514e+06	3.077320e+07	6.309659	645.423295	2155.704545	288.170455	580.889205	103.267045	16
std	1.383952e+06	8.725344e+06	3.873810	536.499990	2013.393716	298.500277	571.632472	103.831374	21
min	1.090000e+05	8.854000e+06	1.000000	7.000000	7.000000	1.000000	1.000000	0.000000	0
25%	2.527000e+05	2.455733e+07	3.000000	230.000000	631.500000	73.000000	163.000000	26.000000	3
50%	6.750000e+05	3.413650e+07	6.000000	518.500000	1583.000000	190.500000	416.000000	70.000000	9
75%	2.250000e+06	3.779200e+07	9.000000	930.000000	3026.000000	405.500000	817.250000	146.250000	23
max	6.200000e+06	4.286600e+07	20.000000	2729.000000	10554.000000	1570.000000	3025.000000	634.000000	142

Comments:

A. Team Salaries and Payroll

- Average Salary: $1.33M
- Average Team Payroll: 30.77M with a wide range from 8.85M to 42.86M.
- 75% of players are paid under ~2.25M, while some players are paid very well (up to $6.2M).

B. Years and Matches

- Average player has 6.3 years of experience, 50% of players have less than 6 years of experience, but the top 25% have 9 years or more.
- Average 645 matches, 50% of players have less than 518 matches, while the top 25% have more than 930 matches.
-

--> Only a few players have a long playing time and maintain a long career.

C. Player Statistics

- Distributions are right-skewed for metrics like at_bats, runs, hits, and homeruns.
- Top 25% clearly outperform in key stats like hits, runs, and home runs.
- Fielding is stable across players; batting and scoring show big performance gaps.

D. Demographic

- Some cities have a mix of all three groups, but most are still overwhelmingly white.

2. Check any records with missing values and handle the missing data as appropriate.

In [21]:
```
# Show the number of missing values in each column
missing_data = df.isnull().sum()

# Filter out columns with missing values
missing_data = missing_data[missing_data > 0]

# Show results
if not missing_data.empty:
    print("Missing Data:")
    print(missing_data)
else:
    print("No missing data found in the dataset.")
```

```
Missing Data:
white_percent_in_city       23
black_percent_in_city       23
hispanic_percent_in_city    23
dtype: int64
```

In [22]:
```
# Handle missing data by filling in the average value of each column
df.fillna({
    "white_percent_in_city": df["white_percent_in_city"].mean(),
    "black_percent_in_city": df["black_percent_in_city"].mean(),
    "hispanic_percent_in_city": df["hispanic_percent_in_city"].mean()
}, inplace=True)

# Check for missing data
missing_data_after = df.isnull().sum()

missing_data_after
```

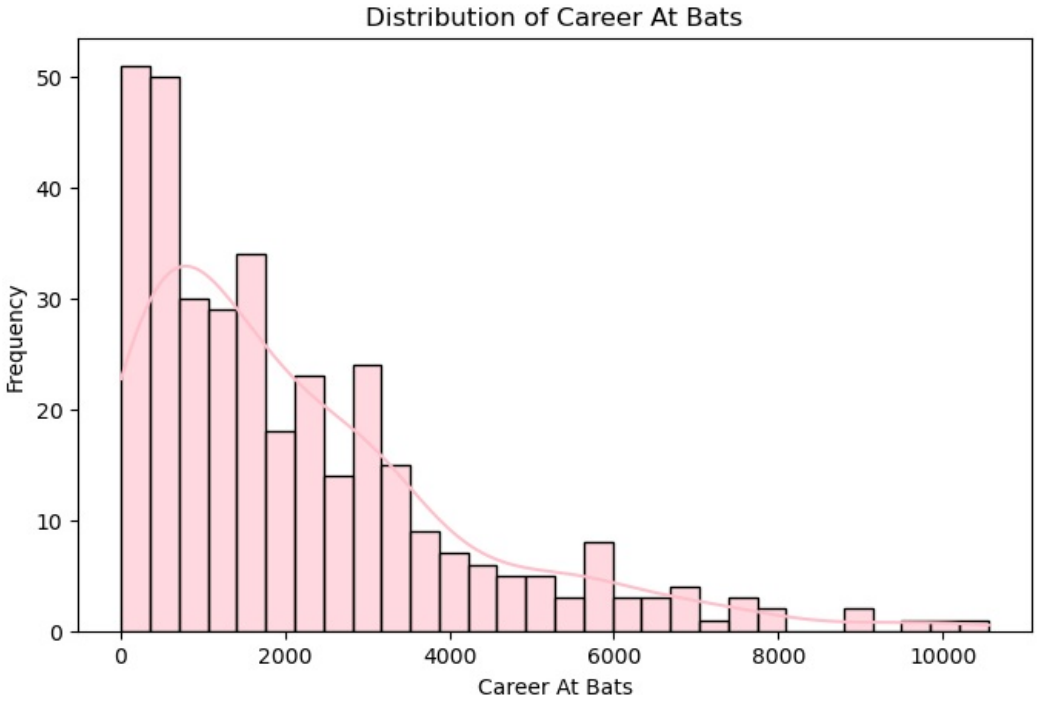
Out[22]:
```
salary                      0
player_position             0
race                        0
team_payroll                0
years_played                0
games_played                0
career_at_bats              0
career_runs                 0
career_hits                 0
career_doubles              0
career_triples              0
career_homeruns             0
career_runs_batted_in       0
career_batting_average      0
career_walks                0
career_strike_outs          0
career_stolen_bases         0
career_fielding_percent     0
white_percent_in_city       0
black_percent_in_city       0
hispanic_percent_in_city    0
dtype: int64
```

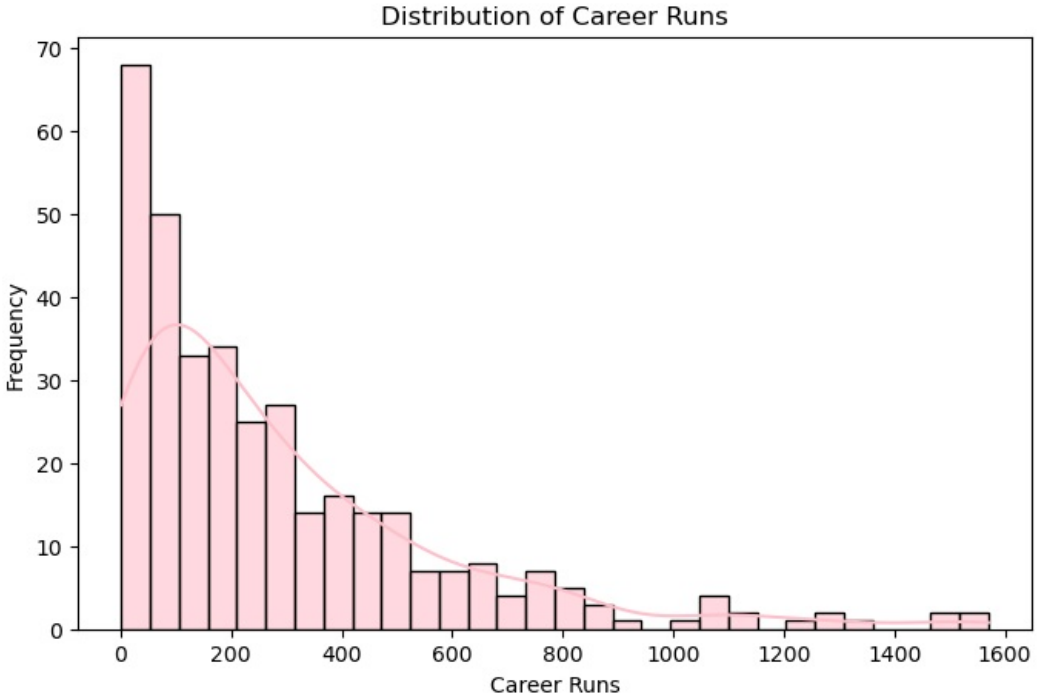
3. Build graphs visualizing the following and comment on the obtained visual insights

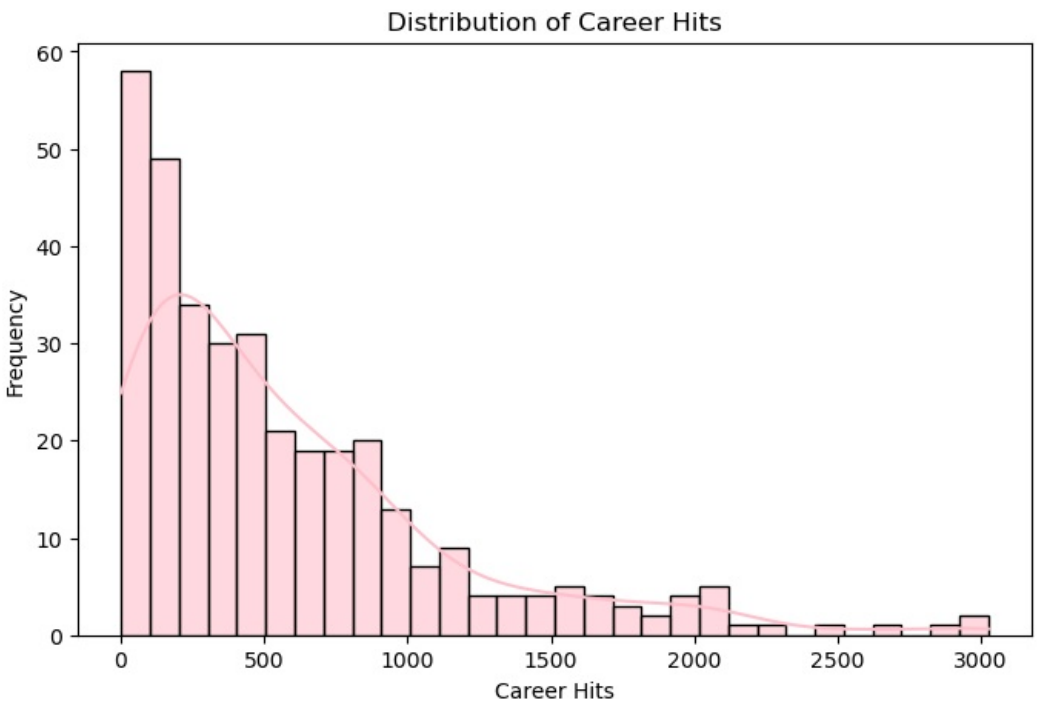
A. The distribution of one or more individual continuous variables

In [25]:
```
import matplotlib.pyplot as plt
import seaborn as sns

continuous_vars = [
    "career_at_bats", "career_runs", "career_hits"
]

# Set up chart
plt.figure(figsize=(12, 8))

# Draw histogram and KDE plot for each variable
for var in continuous_vars:
    plt.figure(figsize=(8, 5))
    sns.histplot(df[var], bins=30, kde=True, color="pink", alpha=0.6)
    plt.title(f"Distribution of {var.replace('_', ' ').title()}")
    plt.xlabel(var.replace('_', ' ').title())
    plt.ylabel("Frequency")
    plt.show()
```

```
<Figure size 1200x800 with 0 Axes>
```


Distribution of Career At Bats
Frequency
Career At Bats
0
10
20
30
40
50
0
2000
4000
6000
8000
10000


Distribution of Career Runs
Frequency
Career Runs
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
1400
1600

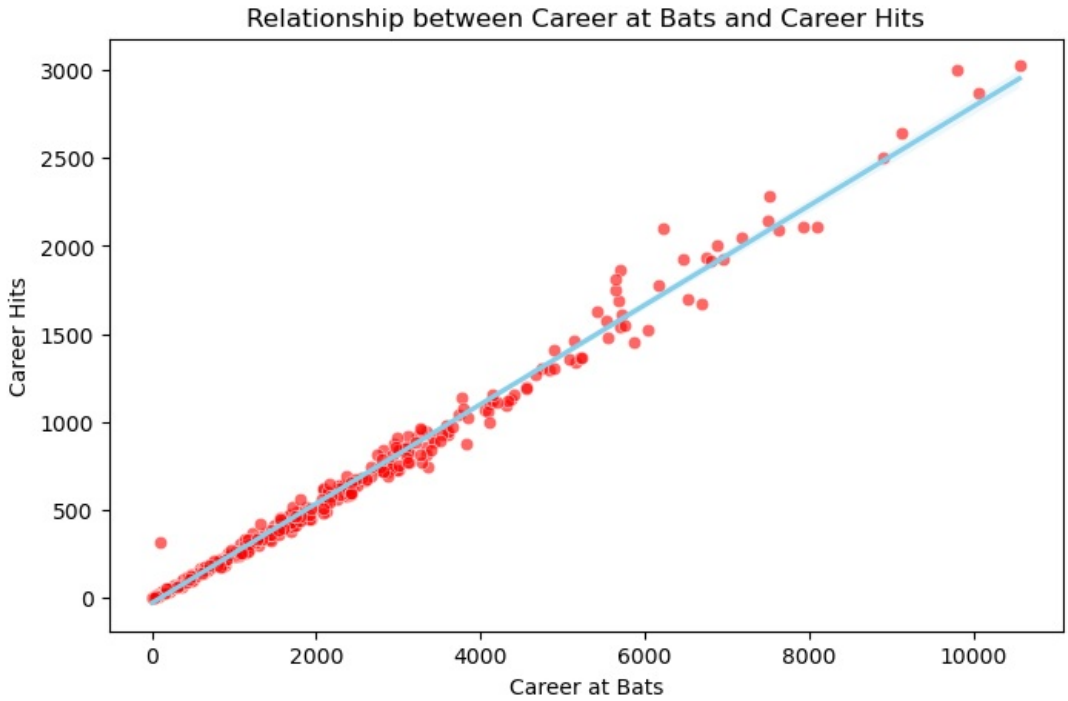


Comments:

- Most of the parameters have a right-skewed distribution, with the majority of players having average or low performance.
- A small number of players have outstanding performance, extending the right tail of the distribution. + Career at Bats suggests that extended playing time is rare and limited to top or long-tenured players + Career Runs reflects a typical performance gap: few high scorers, many average-level players. + Career Hits shows that high consistency and longevity are uncommon in the league.
- This reflects that in professional sports, only a small group of players reach the top, while the majority have average performance.

B. The relationship of a pair of continuous variables.

In [28]:
```
var_x = "career_at_bats"
var_y = "career_hits"

# Draw a scatter plot
plt.figure(figsize=(8, 5))
sns.scatterplot(x=df[var_x], y=df[var_y], alpha=0.6, color="red")
sns.regplot(x=df[var_x], y=df[var_y], scatter=False, color="skyblue")

# Set title and label
plt.title(f"Relationship between Career at Bats and Career Hits")
plt.xlabel(f"Career at Bats")
plt.ylabel(f"Career Hits")
plt.show()
```



Comments:

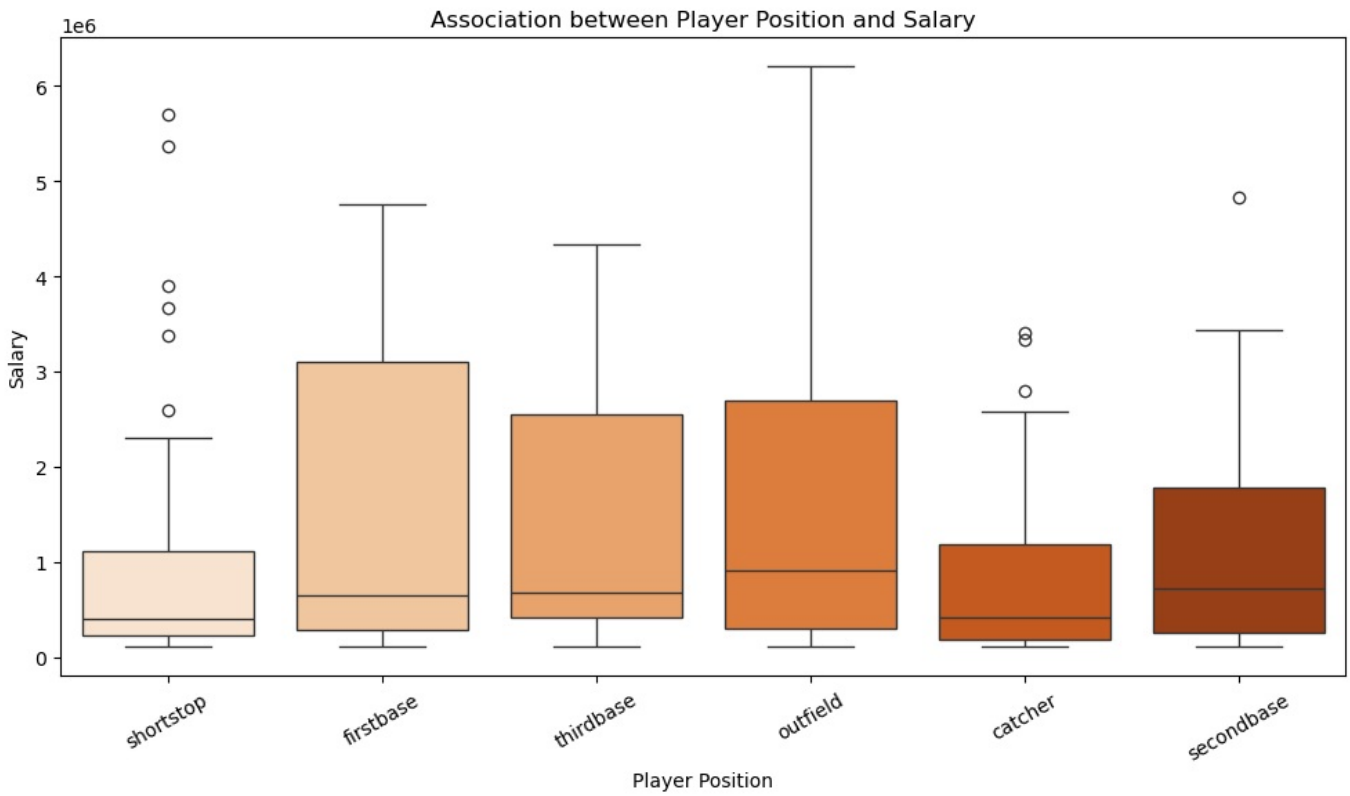
- The relationship between career_at_bats and career_hits shows a strong linear trend.
- The correlation coefficient is highly positive, meaning that more at-bats tend to increase the number of hits.
- The skyblue regression line shows a nearly perfect linear relationship, meaning that players with more at-bats tend to have more hits.
- Some data points are far from the regression line, indicating that there are some outliers – some players may have better or worse than average at-bat rates.

-->There is a clear linear relationship between at-bats and hits, suggesting that more opportunities to play often lead to better performance.

C. The association b/w a categorical variable and a continuous one.

In [31]:
```
cat_var = "player_position"  # Player position (categorical)
cont_var = "salary"  # Salary (continuous)

# Draw a boxplot
plt.figure(figsize=(12, 6))
sns.boxplot(x=df[cat_var], y=df[cont_var], palette="Oranges")

# Set title and label
plt.title(f"Association between Player Position and Salary")
plt.xlabel(f"Player Position")
plt.ylabel(f"Salary")
plt.xticks(rotation=30)
plt.show()
```

```
C:\Users\ACER\AppData\Local\Temp\ipykernel_12096\3695848464.py:6: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(x=df[cat_var], y=df[cont_var], palette="Oranges")
```



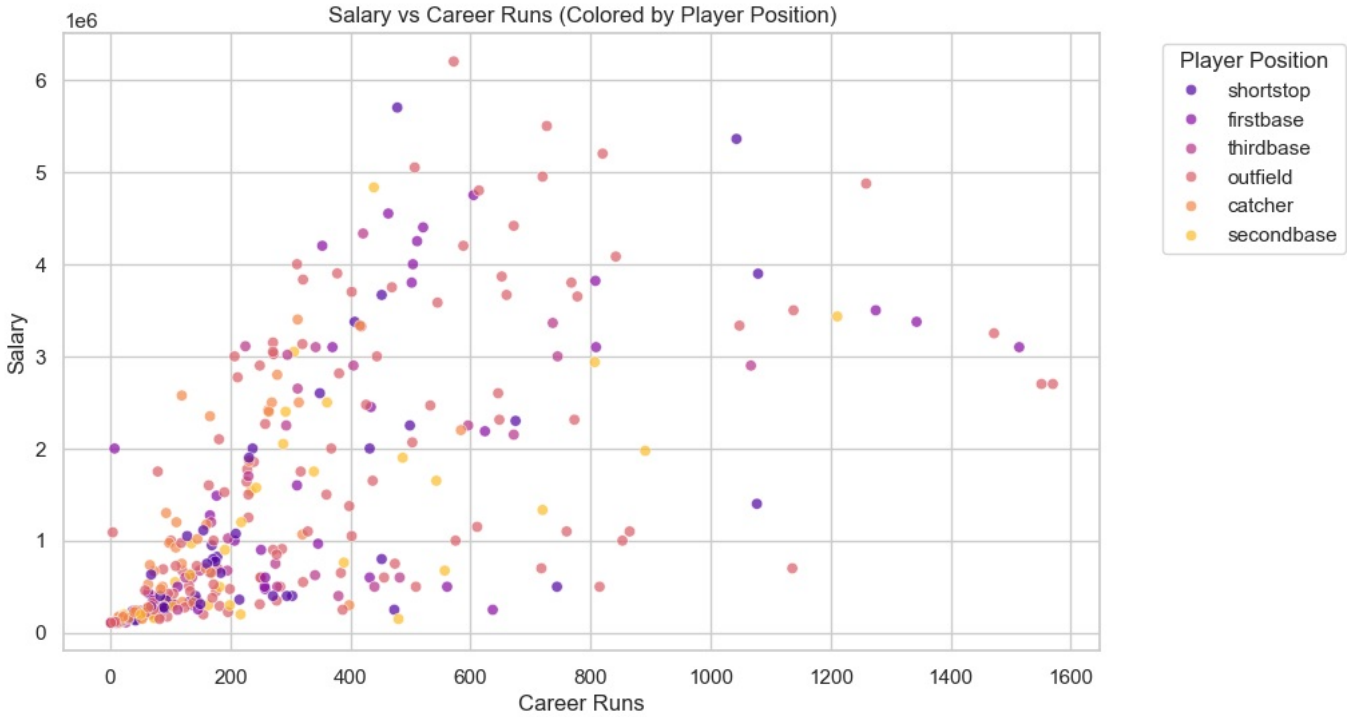
Comments:

- Salaries vary significantly between player positions.
- Some positions such as First Base, Outfield, and Third Base have higher average salaries than others.
- Outliers appear at all positions, indicating that some players earn very high salaries compared to the rest.
- Catcher and Second Base have the lowest average salaries, with a narrower salary range than other positions.

--> Position affects player salaries, with First Base, Outfield, and Third Base tending to be higher paid, while Catcher and Second Base have salaries below average.

D. The relationship between more than two variables, e.g., using semantic mappings.

In [34]:
```
# Set the style for the plots
sns.set(style="whitegrid")

# Scatter Plot with Color Encoding (Salary vs Career Runs, Colored by Player Position)
plt.figure(figsize=(10, 6))
sns.scatterplot(
    x=df['career_runs'],
    y=df['salary'],
    hue=df['player_position'],  # Encode player position using color
    palette='plasma',
    alpha=0.7
)
plt.title('Salary vs Career Runs (Colored by Player Position)')
plt.xlabel('Career Runs')
plt.ylabel('Salary')
plt.legend(title='Player Position', bbox_to_anchor=(1.05, 1), loc='upper left')
plt.show()
```



Comments:

- Players with more career runs tend to have higher salaries.
- Most players are concentrated under 400 runs and $1.5M salary, indicating average performance and pay.
- Outfielders lead in both runs and salary, Firstbase/Thirdbase are mid to high range, Catchers, Secondbase, Shortstops stay mostly low.

--> There is a clear upward trend between career runs and salary, with outfielders standing out as top earners and performers.

4. Display unique values of a categorical variable and their frequencies.

In [37]:
```
cat_var = "player_position"

# Calculate the frequency of occurrence of each value
table = df[cat_var].value_counts()

# Display frequency table as DataFrame
table_df = table.reset_index()
table_df.columns = [cat_var, "Frequency"]

print(table_df)
```

```
  player_position  Frequency
0        outfield        136
1         catcher         52
2       shortstop         49
3       firstbase         45
4      secondbase         36
5       thirdbase         34
```

5. Build a contingency table of two potentially related categorical variables. Conduct a statistical test of the independence between them and interpret the results.

In [39]:
```
import scipy.stats as stats

# Select two categorical variables for the contingency table.
cat_var1 = "player_position"
cat_var2 = "race"

# Create a contingency table
contingency_table = pd.crosstab(df[cat_var1], df[cat_var2])

# Perform Chi-square test to check for independence between two variables
chi2_stat, p_value, dof, expected = stats.chi2_contingency(contingency_table)

# Display results
print("Contingency Table:")
print(contingency_table)
```

```
print("\nChi-Square Test Results:")
print(f"Chi-square Statistic: {chi2_stat:.2f}")
print(f"P-value: {p_value:.4f}")
print(f"Degrees of Freedom: {dof}")

# Interpretation of results
alpha = 0.05  # Significance level
if p_value < alpha:
    print("Conclusion: There is a statistically significant relationship between player position and race (reje
else:
    print("Conclusion: There is not enough evidence to conclude that player position and race are related.")
```

```
Contingency Table:
race             black  hispanic  white
player_position
catcher              2         7     43
firstbase           10         9     26
outfield            72        19     45
secondbase           8         7     21
shortstop            7        19     23
thirdbase            9         3     22

Chi-Square Test Results:
Chi-square Statistic: 76.63
P-value: 0.0000
Degrees of Freedom: 10
Conclusion: There is a statistically significant relationship between player position and race (rejecting the in
dependence hypothesis).
```

Intepret the results:

- Since p-value = 0.0000 < 0.05, we reject the independence hypothesis.
- This means that there is a statistically significant relationship between player position and race.
- Certain positions are more frequently occupied by players from specific racial groups. For example:
  - Outfield has the highest number of Black players (72).
  - Catcher is predominantly White (43).
  - Shortstop has a relatively high number of Hispanic players (19).

6. Retrieve one or more subset of rows based on two or more criteria and present descriptive statistics on the subset(s).

In [42]:
```
# Define new criteria: Select players who play in "firstbase", have more than 100 home runs, and earn more than
subset_criteria_extended = (df["player_position"] == "firstbase") & (df["career_homeruns"] > 100) & (df["salary

# Extract the subset of data based on the new criteria
subset_df_extended = df[subset_criteria_extended]

# Generate descriptive statistics for the new subset
subset_statistics_extended = subset_df_extended.describe()

# Print the results
print("**Descriptive statistics for the subset of players in 'firstbase' position, with more than 100 home runs
print(subset_statistics_extended)
```

```
**Descriptive statistics for the subset of players in 'firstbase' position, with more than 100 home runs and a s
alary above $3 million**:
             salary  team_payroll  years_played  games_played  career_at_bats  \
count  1.100000e+01  1.100000e+01     11.000000     11.000000       11.000000
mean   3.845000e+06  3.515623e+07     10.727273   1394.000000     5199.000000
std    5.379359e+05  6.244453e+06      4.540725    643.478049     2574.666542
min    3.100000e+06  2.455733e+07      7.000000    696.000000     2297.000000
25%    3.437500e+06  3.076672e+07      7.000000    950.500000     3364.500000
50%    3.820000e+06  3.594817e+07      9.000000   1028.000000     3780.000000
75%    4.225000e+06  3.977656e+07     13.500000   1699.500000     6581.500000
max    4.750000e+06  4.286600e+07     20.000000   2562.000000     9789.000000

       career_runs  career_hits  career_doubles  career_triples  \
count    11.000000    11.000000       11.000000       11.000000
mean    795.000000  1511.818182      283.909091       33.272727
std     400.718854   816.594981      166.496519       40.593327
min     353.000000   590.000000       91.000000        3.000000
25%     507.500000   892.000000      161.000000       13.000000
50%     605.000000  1139.000000      222.000000       15.000000
75%    1042.000000  2017.500000      384.000000       33.500000
max    1514.000000  3005.000000      634.000000      134.000000

       career_homeruns  career_runs_batted_in  career_batting_average  \
count        11.000000              11.000000               11.000000
mean        215.272727             831.363636              284.363636
std          82.072030             382.700738               21.341381
min         123.000000             473.000000              247.000000
25%         161.500000             589.000000              270.500000
50%         191.000000             636.000000              286.000000
75%         239.000000             931.500000              302.000000
max         414.000000            1562.000000              311.000000

       career_walks  career_strike_outs  career_stolen_bases  \
count     11.000000           11.000000            11.000000
mean     606.090909          700.454545            81.727273
std      285.326639          240.264589           122.156531
min      287.000000          343.000000             0.000000
25%      402.500000          606.000000            21.000000
50%      527.000000          697.000000            33.000000
75%      742.500000          788.500000            70.000000
max     1147.000000         1224.000000           412.000000

       career_fielding_percent  white_percent_in_city  black_percent_in_city  \
count                 11.000000              11.000000              11.000000
mean                983.727273              76.703784              13.863774
std                  17.418903               8.090266               6.673438
min                 940.000000              67.668960               3.741786
25%                 980.000000              72.601955               9.049699
50%                 992.000000              74.619650              12.801390
75%                 993.500000              80.655090              17.528360
max                 995.000000              94.696270              26.153940

       hispanic_percent_in_city
count                 11.000000
mean                   9.432442
std                    7.336604
min                    1.280389
25%                    2.424793
50%                   10.819810
75%                   15.566665
max                   20.090960
```

Comments:

- "Firstbase" players with more than 100 home runs and earning over 3 million generally have long careers (10 years) and high salaries ($3.85 million).
- They exhibit strong batting performance, averaging over 200 home runs and more than 1,500 career hits.
- City demographics suggest they mostly come from areas with a high White population (~77%)

7. Conduct a statistical test of the significance of the difference between the means of two subsets of the data and interpret the results.

In [45]:
```python
import scipy.stats as stats

# Define two subsets for comparison
subset1 = df[(df["player_position"] == "firstbase") & (df["career_homeruns"] > 100)]["salary"]
subset2 = df[(df["player_position"] == "outfield") & (df["career_homeruns"] > 100)]["salary"]

# Conduct an independent t-test to compare the means of the two subsets
t_stat, p_value = stats.ttest_ind(subset1, subset2, equal_var=False)
```

```
# Display results
print("**Statistical Test for Salary Difference Between 'Firstbase' and 'Outfield' Players with More than 100 H
print(f"T-statistic: {t_stat:.2f}")
print(f"P-value: {p_value:.4f}")

# Interpretation
alpha = 0.05  # Significance level
if p_value < alpha:
    print(" Conclusion: There is a statistically significant difference in salaries between 'Firstbase' and 'Ou
else:
    print(" Conclusion: There is no significant difference in salaries between 'Firstbase' and 'Outfield' playe
```

```
**Statistical Test for Salary Difference Between 'Firstbase' and 'Outfield' Players with More than 100 Home Runs
**
T-statistic: 0.81
P-value: 0.4251
 Conclusion: There is no significant difference in salaries between 'Firstbase' and 'Outfield' players with more
than 100 home runs.
```

Comments:

- Since P-value (0.4251) > 0.05, we fail to reject the null hypothesis.
- This means that there is no statistically significant difference in salaries between "Firstbase" and "Outfield" players with more than 100 home runs.
- The difference observed in mean salaries could be due to random variation rather than a true systematic difference.

8. Create one or more tables that group the data by a certain categorical variable and display summarized information for each group (e.g., the mean or sum within the group).

In [48]:
```
# Group the data by player position and summarize key statistics
grouped_stats = df.groupby("player_position").agg(
    mean_salary=("salary", "mean"),
    total_salary=("salary", "sum"),
    mean_home_runs=("career_homeruns", "mean"),
    total_home_runs=("career_homeruns", "sum"),
    mean_games_played=("games_played", "mean"),
    total_games_played=("games_played", "sum")
).reset_index()

# Display the summarized table
print("**Grouped Summary by Player Position**")
print(grouped_stats)
```

```
**Grouped Summary by Player Position**
  player_position   mean_salary  total_salary  mean_home_runs  \
0         catcher  8.925192e+05      46411001       26.326923
1       firstbase  1.586781e+06      71405167       84.911111
2        outfield  1.539324e+06     209348100       70.625000
3      secondbase  1.170208e+06      42127499       28.083333
4       shortstop  1.069211e+06      52391324       26.795918
5       thirdbase  1.382647e+06      47010000       61.058824

   total_home_runs  mean_games_played  total_games_played
0             1369         426.211538               22163
1             3821         702.511111               31613
2             9605         706.044118               96022
3             1011         677.500000               24390
4             1313         632.918367               31013
5             2076         646.705882               21988
```

Comments:

- Firstbase players tend to be among the highest-paid and most productive in terms of home runs.
- Outfielders play the most games and contribute heavily to home run totals.
- Catchers have lower salaries and play fewer games, reflecting the physically demanding nature of their position.

In [50]:
```
# Group the data by race and summarize key statistics
grouped_stats_race = df.groupby("race").agg(
    mean_salary=("salary", "mean"),
    total_salary=("salary", "sum"),
    mean_home_runs=("career_homeruns", "mean"),
    total_home_runs=("career_homeruns", "sum"),
    mean_games_played=("games_played", "mean"),
    total_games_played=("games_played", "sum")
).reset_index()

# Print the second grouped summary table
```

```
print("**Grouped Summary by Race**")
print(grouped_stats_race)
```

```
**Grouped Summary by Race**
       race   mean_salary  total_salary  mean_home_runs  total_home_runs  \
0     black  1.654350e+06     178669850       68.231481             7369
1  hispanic  1.050723e+06      67246267       33.140625             2121
2     white  1.237650e+06     222776974       53.916667             9705

   mean_games_played  total_games_played
0         730.972222               78945
1         523.109375               33479
2         637.583333              114765
```

Comments:

- Black players tend to have the highest salaries and play the most games on average.
- White players have the highest total home runs and total games played, showing a strong long-term presence.
- Hispanic players have the lowest salary, home runs, and games played, suggesting they might have shorter careers or different roles within teams.

9. Implement a linear regression model and interpret its output including its accuracy

In [83]:
```
import statsmodels.api as sm

# Select columns as independent variables and the target variable
selected = ['years_played', 'career_hits', 'career_runs']
target = 'career_triples'

# Drop rows with NaN values
df = df[selected + [target]].dropna()

# Create X (features) and y (target)
X_full = sm.add_constant(df[selected])  # Add an intercept
y = df[target]

# Build an OLS model with all variables
model_full = sm.OLS(y, X_full).fit()

print(" Full model summary:")
print(model_full.summary())

print("\nP-values (Full model):")
print(model_full.pvalues)
```

```
 Full model summary:
                            OLS Regression Results
==============================================================================
Dep. Variable:         career_triples   R-squared:                       0.742
Model:                            OLS   Adj. R-squared:                  0.739
Method:                 Least Squares   F-statistic:                     333.0
Date:                Thu, 27 Mar 2025   Prob (F-statistic):          6.71e-102
Time:                        23:04:08   Log-Likelihood:                -1339.0
No. Observations:                 352   AIC:                             2686.
Df Residuals:                     348   BIC:                             2701.
Df Model:                           3
Covariance Type:            nonrobust
==================================================================================
                     coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------
const             -1.4039      1.230     -1.141      0.255      -3.824       1.016
years_played      -0.1415      0.321     -0.441      0.659      -0.772       0.489
career_hits        0.0162      0.004      3.698      0.000       0.008       0.025
career_runs        0.0328      0.007      4.501      0.000       0.018       0.047
==============================================================================
Omnibus:                      124.508   Durbin-Watson:                   1.973
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             1335.343
Skew:                           1.146   Prob(JB):                    1.08e-290
Kurtosis:                      12.262   Cond. No.                     1.96e+03
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 1.96e+03. This might indicate that there are
strong multicollinearity or other numerical problems.

P-values (Full model):
const           0.254625
years_played    0.659258
career_hits     0.000252
career_runs     0.000009
dtype: float64
```

In [79]:

```python
# Function for backward elimination
def build_refined_model(X, y, significance_level=0.05):
    while True:
        model = sm.OLS(y, X).fit()
        p_values = model.pvalues[1:]  # Exclude the intercept's p-value

        # Check if all p-values are below the significance level
        if all(p_values < significance_level):
            return model  # Return the model when no variable exceeds the significance threshold

        # Identify the feature with the highest p-value
        worst_feature = p_values.idxmax()
        print(f"Removing feature '{worst_feature}' with p-value = {p_values.max():.4f}")

        # Drop that feature from X
        X = X.drop(columns=[worst_feature])

# Apply backward elimination
refined_model = build_refined_model(X_full, y)
print("\n Refined model summary (after backward elimination):")
print(refined_model.summary())

# Compare Adjusted R²
print(f"\nAdjusted R² (Full): {model_full.rsquared_adj:.4f}")
print(f"Adjusted R² (Refined): {refined_model.rsquared_adj:.4f}")
```

```
Removing feature 'years_played' with p-value = 0.6593

 Refined model summary (after backward elimination):
                            OLS Regression Results
==============================================================================
Dep. Variable:         career_triples   R-squared:                       0.742
Model:                            OLS   Adj. R-squared:                  0.740
Method:                 Least Squares   F-statistic:                     500.6
Date:                Thu, 27 Mar 2025   Prob (F-statistic):          2.92e-103
Time:                        23:03:36   Log-Likelihood:                -1339.1
No. Observations:                 352   AIC:                             2684.
Df Residuals:                     349   BIC:                             2696.
Df Model:                           2
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const         -1.8043      0.830     -2.174      0.030      -3.437      -0.172
career_hits    0.0153      0.004      4.026      0.000       0.008       0.023
career_runs    0.0331      0.007      4.555      0.000       0.019       0.047
==============================================================================
Omnibus:                      124.422   Durbin-Watson:                   1.974
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             1306.194
Skew:                           1.152   Prob(JB):                    2.31e-284
Kurtosis:                      12.151   Cond. No.                     1.30e+03
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 1.3e+03. This might indicate that there are
strong multicollinearity or other numerical problems.

Adjusted R² (Full): 0.7394
Adjusted R² (Refined): 0.7400
```

Comments:

1. Model Fit

R-squared = 0.742: About 74.2% of the variation in career_triples is explained by the model. Adjusted R-squared = 0.740: The model remains strong after adjusting for the number of predictors, indicating good overall fit without overfitting.

2. Coefficients and Significance

Both predictors (career_hits and career_runs) have very low p-values (< 0.001) → statistically significant.

3. Other Model Diagnostics

F-statistic = 500.6 with p < 0.001 → The overall model is highly significant. Durbin-Watson = 1.974 → Residuals show no sign of autocorrelation. Condition Number = 1300+ → Slight risk of multicollinearity, but still acceptable.

--> This is a strong and reliable model. It shows that career_hits, years_played and career_runs both have a meaningful impact on the number of career triples. The model explains a large part of the variation and passes important statistical tests, making it suitable for analysis and interpretation.

Conclusion

This project provides a comprehensive overview of how to explore and analyze data sets using various statistical and visualization methods. Starting with descriptive statistics, I was able to understand the general structure and variability in the data. I carefully handled missing values to ensure that the analysis remained reliable. Visualizations such as histograms, scatter plots, and correlation heat maps helped to clearly demonstrate key patterns and relationships.

I also used statistical tests to test the significance of differences between groups and built simple regression models to determine which variables had the strongest impact on other variables.

Working on this project helped me realize both the possibilities and the difficulties of working with real-world data. I may not have chosen the best techniques every time, but each step taught me something new. It reminded me that data analysis is not just about using the right tools, but also about asking the right questions, thinking clearly, and continuing to improve.

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